Latent Sequence Drift for Influenza Surveillance

Representation Learning Validates Genetic Drift but Does Not Yet Improve Hospitalisation Forecasting

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1 Problem Formulation and Motivation

Seasonal influenza imposes a large and highly variable burden on healthcare systems worldwide, causing an estimated 290,000–650,000 respiratory deaths per year (Iuliano et al. 2018; Krammer et al. 2018); in the United States alone, FluSurv-NET hospitalisation rates have ranged from 8.7 to 126.2 per 100,000 across the seventeen seasons from 2009–10 to 2025–26, a more than fourteen-fold difference. This variability makes resource planning difficult: a season that turns out to be severe may overwhelm hospital capacity and antiviral stockpiles, while an overly cautious allocation in a mild season wastes scarce public health funding (Chaiyakunapruk et al. 2017).

A genomic early-warning signal could address this challenge directly. Influenza haemagglutinin (HA) sequences are deposited in public databases within weeks of sample collection, well before the hospitalisation peak (Shu and McCauley 2017; Squires et al. 2012); hence, if the degree of antigenic novelty encoded in those sequences correlates with downstream disease burden, a model that extracts a drift score from the sequence data could provide an earlier indication of season severity than purely retrospective surveillance measures such as ILI % or FluSurv-NET rates.

The core idea of my project is to chain three components into a single pipeline: (i) a variational autoencoder (VAE) trained on HA nucleotide sequences to learn a continuous latent representation without phylogenetic or antigenic supervision, (ii) a drift score computed as the Euclidean distance between consecutive season centroids in latent space, and (iii) a SIRC compartmental model (Casagrandi et al. 2006) that translates the drift score into a predicted hospitalisation rate via a mechanistic immunity-waning parameter. I compare a Conv1D VAE against a Transformer VAE (Vaswani et al. 2017), and benchmark both against a PCA baseline and simple historical-average predictors.
The project addresses three research questions:

RQ1. Do VAE latent spaces capture biologically meaningful seasonal drift, as measured by concordance with pairwise Hamming genetic distance?

RQ2. Does incorporating sequence-derived drift into a SIRC model improve seasonal hospitalisation forecasting beyond simple baselines?

RQ3. Which of the two autoencoder architectures — Conv1D VAE or Transformer VAE — offers the better trade-off between representation quality, computational cost, and downstream utility?

The main result is mixed in a useful way. The representation learning objective is met: the Conv1D VAE recovers drift signals that correlate significantly with Hamming distance for both H1N1 and H3N2 ($\rho \approx 0.60$, $p < 0.05$), a result not replicated by PCA. Conversely, the forecasting objective is not met: the VAE-SIRC pipeline does not beat a historical-average baseline, while PCA-SIRC paradoxically achieves the best forecast despite failing biological validation. This pattern identifies the SIRC calibration stage, not the learned representation, as the binding constraint on predictive performance — a finding that points directly to the model structures and covariates needed for future progress.

2 Professional Ambitions Reflection

My professional ambition lies at the intersection of machine learning and public health analytics: developing computational tools that translate high-dimensional biomedical data into actionable inputs for resource allocation, cost-effectiveness analysis, and health policy. Health economics is fundamentally concerned with optimising the deployment of scarce resources—hospital beds, vaccine doses, antiviral stockpiles—under uncertainty about future disease burden (Chaiyakunapruk et al. 2017); seasonal influenza exemplifies this challenge acutely, since substantial inter-annual variation in hospitalisation rates means that static budget

models systematically misallocate resources, and a tool providing earlier, sequence-informed signals of season severity would have direct economic value.

This project connects to that ambition concretely. The pipeline I constructed—from HA nucleotide sequences through VAE-derived latent drift scores to SIRC-predicted hospitalisation rates—is precisely the type of translational modelling that sits at the boundary between computational biology and health economics. Building it required engaging with biological sequence data at a level of detail I had not previously encountered, reinforcing my conviction that impactful clinical machine learning demands genuine domain fluency, not merely technical competence; the latent drift metric, if validated against serological data, could serve as an early-warning input to cost-effectiveness models evaluating vaccination timing and breadth.

The architectural comparison between convolutional and Transformer VAEs is itself a health-economic question in miniature: the Transformer requires approximately ten times the wall-clock training time per epoch, and if it offers only marginal improvement in drift quantification, the additional cost is not justified for routine surveillance—a trade-off that mirrors the resource-constraint reasoning endemic to health economics. Learning to frame technical choices in cost-effectiveness terms is a skill I intend to carry into future work in clinical risk prediction and public health modelling.

More broadly, this project has deepened my understanding of how modern deep learning architectures can be applied to sequential biological data in a way that is both scientifically rigorous and policy-relevant. Integrating unsupervised representation learning with mechanistic compartmental models has reinforced my conviction that impactful health-economic research will increasingly require fluency in machine learning—not as black-box predictors, but as tools for extracting interpretable, low-dimensional summaries from complex biological systems that can be embedded within the causal frameworks upon which policy decisions depend; in the longer term, I aim to apply these skills to real-time genomic surveillance systems where sequence data arrive continuously, models must be updated incrementally, and outputs must be communicated to public health professionals who are not machine learning specialists.

3 Data Understanding and Preparation

3.1 Data Sources

The pipeline draws on four distinct data streams, each chosen because it measures a complementary facet of annual influenza activity.

NCBI Influenza Research Database (IRD). I downloaded all available haemagglutinin (HA) nucleotide sequences for influenza A H1N1pdm09 and H3N2 from the NCBI IRD (Squires et al. 2012; Shu and McCauley 2017), restricting to human samples collected between 2009 and 2026. After quality filtering and alignment with MAFFT, the global corpus contains 31,222 H1N1 sequences (trimmed to 1,759 nt) and 46,656 H3N2 sequences (trimmed to 1,854 nt). For the drift analysis I further restrict to USA-collected sequences, yielding 25,760 H1N1 sequences (79.4 % of the global pool) and 38,109 H3N2 sequences (76.8 %). The USA subset is large enough to compute stable per-season centroids while remaining directly comparable to USA-based surveillance targets.

FluSurv-NET (CDC). FluSurv-NET provides laboratory-confirmed influenza hospitalisation rates per 100,000 population for each flu season, aggregated across a sentinel network of hospitals. I use the all-age, all-subtype overall rate as the downstream prediction target; across the 17 seasons from 2009–10 to 2025–26, the rate ranges from 8.7 (2011–12) to 126.2 (2024–25) per 100,000, reflecting considerable inter-season variability that a drift-informed model should explain.

ILINet (CDC). ILINet supplies 1,483 weekly observations of weighted influenza-like illness as a percentage of outpatient visits from 1997 to 2024. I use ILI % primarily for exploratory analysis and to contextualise the FluSurv-NET trend; it is not used directly as a model target because it confounds severity with healthcare-seeking behaviour.

WHO/NREVSS Clinical and Public Health Laboratories. WHO/NREVSS reports weekly subtype-positive proportions (H1N1, H3N2, influenza B); I aggregate these to season-level proportions to construct the blended drift score (Section 4.3), so that seasons dominated by H3N2 are weighted toward H3N2 drift rather than averaging the two subtypes equally.

Table 1 summarises all four sources, and Figure 1 shows the four-panel surveillance overview.

Table 1: Summary of all data sources used in this project.

Source	Unit	Date range	Strain split	Key preprocessing
NCBI/IRD	HA sequences (global)	2009–2026	H1N1: 31,222 / H3N2: 46,656	MAFFT alignment, one-hot
NCBI/IRD	HA sequences (USA)	2009–2026	H1N1: 25,760 / H3N2: 38,109	Season-labelled, drift analysis
FluSurv-NET (CDC)	Season hosp. rate	2009–2026	Overall (all ages)	Rates per 100,000
ILINet (CDC)	Weekly ILI %	1997–2026	National	Weighted ILI %
WHO/NREVSS	Weekly subtype typing	2015–2026	H1N1/H3N2/B	Season proportions

3.2 Why These Data Fit the Task

The HA gene is the primary determinant of antigenic novelty: it encodes the surface glycoprotein against which neutralising antibodies are directed, so nucleotide-level changes in HA translate directly into immune escape and population susceptibility (Smith et al. 2004; Petrova and Russell 2018). Influenza remains a leading cause of respiratory mortality worldwide (Iuliano et al. 2018; Krammer et al. 2018), and improving seasonal forecasts has been identified as a high-priority public health objective (Reich et al. 2019); using the full coding sequence rather than just the receptor-binding site lets the autoencoder discover relevant structure without hand-crafting antigenic sites.

Hospitalisation rate is a stronger severity signal than ILI %, because it is less sensitive to changes in healthcare-seeking behaviour or testing intensity; I therefore use FluSurv-NET as the downstream calibration target for the SIRC compartmental model and treat ILI % as supplementary context only.

Restricting the genetic data to USA sequences ensures that the seasonal centroids are comparable to the USA surveillance targets; the USA contributes the majority of HA deposits in the Northern Hemisphere, so the restriction does not severely reduce sample sizes.

3.3 Preprocessing Pipeline

Sequence filtering and alignment. I retain sequences with complete HA records, confirmed human host, and subtype annotation; raw sequences are aligned with MAFFT (Katoh and Standley 2013) using the FFT-NS-2 strategy and then trimmed to a fixed reference length (1,759 nt for H1N1; 1,854 nt for H3N2). H1N1 and H3N2 are processed in entirely separate pipelines throughout to avoid cross-subtype contamination.

Figure 1: Four-panel CDC surveillance overview. *Top-left:* weekly ILI % from ILINet (1997–2024). *Top-right:* season-level subtype proportions (H1N1/H3N2/B). *Bottom-left:* FluSurv-NET overall hospitalisation rates per 100,000 for the 17 seasons under study (range 8.7–126.2). *Bottom-right:* CDC end-of-season vaccine coverage (37–45 %). The inter-season variability in hospitalisation rates is the primary signal I aim to predict.

One-hot encoding. Each nucleotide position is encoded as a five-dimensional indicator vector over the alphabet {A, C, G, T, N}, where the fifth channel absorbs ambiguity codes; the resulting tensor has shape (batch $\times$ 5 $\times$ L) where L is the trimmed sequence length. I use five channels rather than four so that ambiguous bases degrade gracefully rather than forcing arbitrary imputation.

Season assignment. I assign each sequence to the Northern Hemisphere flu season (e.g. "2022–23") using its collection date: months October through April map to the season that *begins* in the preceding October. Sequences collected outside the October–April window are retained for model training but excluded from the season-centroid computation; this assignment captures 89.4 % of H1N1 and 88.8 % of H3N2 sequences, consistent with the strong Oct–Apr collection peak visible in Figure 4.

Train/test split. For the random (non-temporal) evaluation of reconstruction quality, I use a stratified 80/20 split: H1N1 24,977 train / 6,245 test; H3N2 37,324 train / 9,332 test. For temporal evaluation I split by collection year, holding out sequences from 2020 onward as a test set (H1N1 temporal-test: 17,739; H3N2 temporal-test: 27,512); this cutoff is chosen so that the model never sees post-2020 sequences during training, preventing any leakage of future antigenic information into the learned representation.

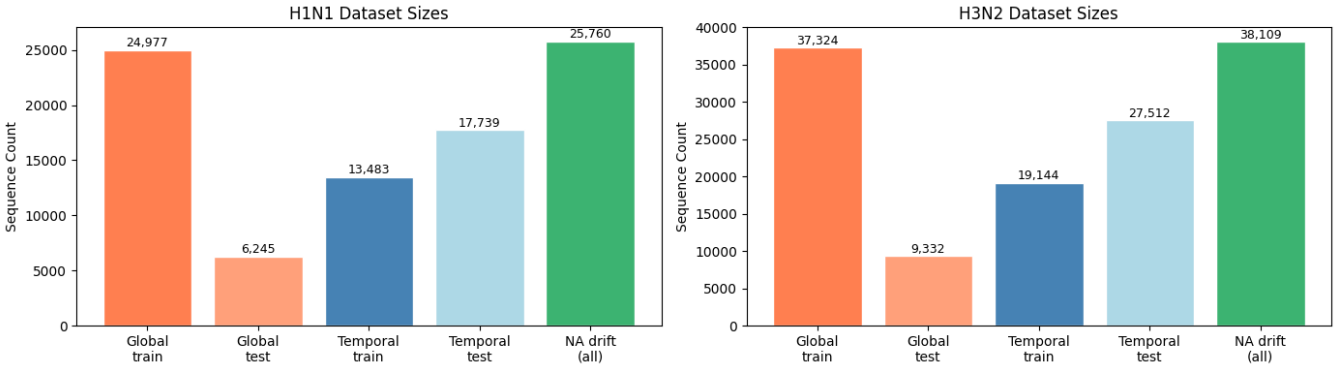

Figure 2: Dataset split sizes for H1N1 and H3N2 across the five subsets used in this project (random train/test, temporal train/test, and the USA-drift analysis set). H3N2 is consistently larger than H1N1, reflecting its longer dominance in Northern Hemisphere circulation.

3.4 Exploratory Data Analysis

Figure 3 shows global and North America deposit counts by year; sequencing effort increases substantially after 2015, which means early-season centroids are computed from fewer sequences and may be less stable. Figure 4 confirms the strong Oct–Apr seasonality in USA deposits, validating the season-assignment heuristic.

Small-n caution. After restricting to seasons 2009–10 through 2025–26 and excluding 2020–21 (COVID-19 disruption), only **15 season-level observations** remain for SIRC calibration and evaluation. Even after leaving out two further seasons for held-out testing, the working sample for correlation analysis is 13 points; this is the single most important limitation of the downstream modelling, as any Pearson or Spearman statistic computed over 13 observations will have wide confidence intervals, and overfitting a three-parameter SIRC model to this series is a genuine risk. I address this explicitly in Section 6 and restrict the SIRC to biologically motivated parameters rather than unrestricted optimisation.

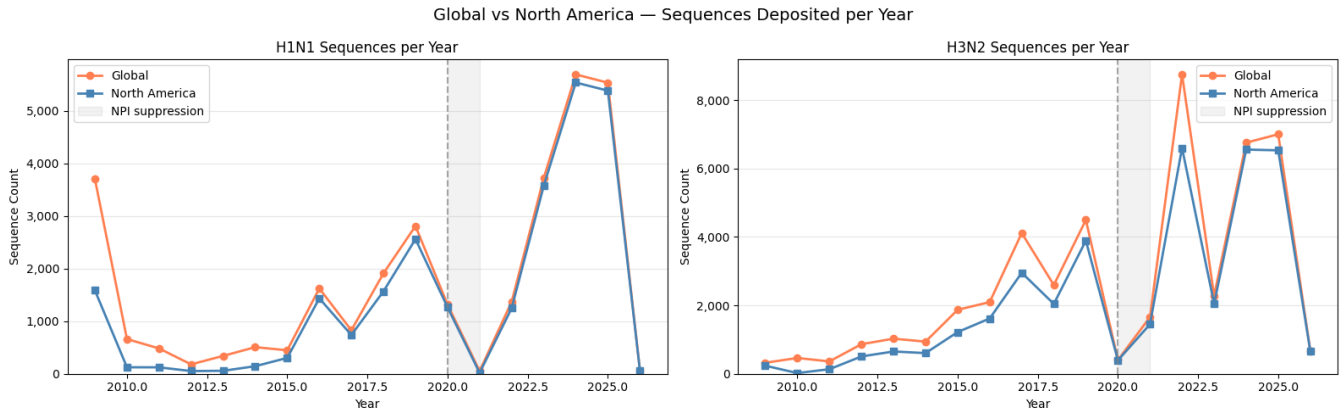

Figure 3: HA sequences deposited per year, stratified by global versus North America. The sharp rise after 2015 reflects expanded genomic surveillance capacity; earlier seasons have sparser coverage and therefore noisier per-season centroids.

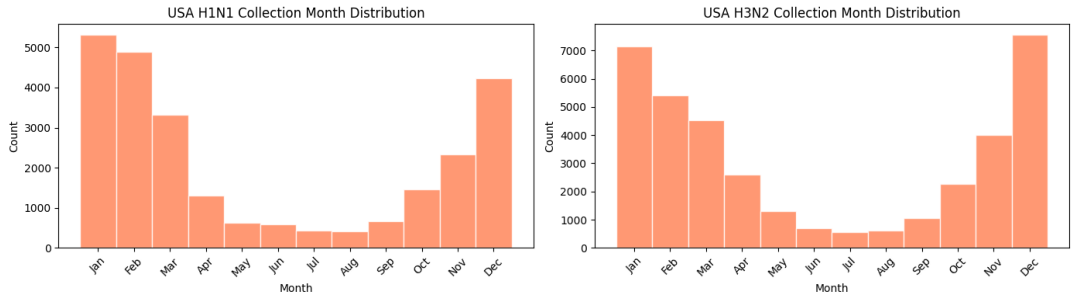

Figure 4: USA sequence collection month histogram. The October–April dominance (89.4 % of H1N1, 88.8 % of H3N2) validates the season-assignment heuristic and confirms that the genetic data tracks the active transmission window.

4 Methodology

4.1 Overall Pipeline

The end-to-end pipeline has five stages. First, HA nucleotide sequences are one-hot encoded and fed into a generative latent model (Conv1D VAE or Transformer VAE) trained on the full multi-year corpus. Second, the trained encoder maps every sequence in the temporal USA subset to a low-dimensional latent vector; these vectors are aggregated to per-season centroids. Third, the L2 distance between consecutive season centroids is computed to produce a raw drift score, which is normalised with an expanding-window min-max transform and blended across subtypes using CDC subtype proportions. Fourth, the drift scores are validated biologically by correlating them with pairwise Hamming distances between season consensus sequences, and structurally by inspecting latent-space geometry via UMAP and silhouette scores. Fifth, the normalised blended drift score is fed into a SIRC compartmental model as a proxy for immunity waning, which is then calibrated against FluSurv-NET hospitalisation rates to evaluate whether latent drift improves downstream epidemiological prediction.

4.2 Models

4.2.1 Conv1D Variational Autoencoder

The Conv1D VAE is the primary model because convolutional filters are well suited to capturing local sequence motifs — short stretches of functionally conserved or rapidly mutating nucleotides — without requiring the quadratic memory of full attention (Kingma and Welling 2014). Recent work has demonstrated that deep generative models can learn biologically meaningful representations of protein and viral sequences without explicit supervision (Riesselman, Ingraham, and Marks 2018; Sinai et al. 2017; Hie et al. 2021); a VAE is appropriate for drift measurement because the continuous, regularised latent space allows centroids to be computed and compared across seasons via simple arithmetic, whereas a deterministic autoencoder would risk a fragmented embedding where distances are not meaningful.

The encoder applies three 1-D convolutional layers over the one-hot input (shape $5 \times L$), followed by global average pooling and a linear projection to the parameters μ and $\log \sigma^2$ of the approximate posterior $q_\phi(\mathbf{z} \mid \mathbf{x})$. The decoder mirrors this structure, upsampling the sampled latent vector back to the original sequence length with transposed convolutions and a per-position softmax over the five-channel alphabet.

The training objective is the standard β-VAE evidence lower bound (Kingma and Welling 2014):

$$\mathcal{L} = \underbrace{-\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} \mid \mathbf{z})]}_{\text{per-position cross-entropy}} + \beta \underbrace{D_{\mathrm{KL}}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big)}_{\text{KL regularisation}}. \tag{1}$$

I use per-position cross-entropy rather than binary cross-entropy because nucleotide labels are four-way categorical, not binary. To prevent posterior collapse — the well-known failure mode of VAEs where the encoder ignores the data and collapses to the prior (Bowman et al. 2016) — I apply linear KL annealing: β is ramped from 0 to its target value over the first 20 epochs. Mixed-precision training (AMP) and early stopping on validation reconstruction loss are used throughout.

4.2.2 Transformer VAE

The Transformer VAE replaces convolutional layers with a multi-head self-attention encoder (Vaswani et al. 2017) to capture long-range dependencies between distant positions in the HA sequence. Long-range coevolution of residues is known to be functionally relevant in influenza HA (Smith et al. 2004), so a global-attention architecture is a principled alternative to the local-context Conv1D model.

The encoder tokenises the one-hot sequence into non-overlapping patches, projects them to a d_{model}-dimensional embedding space, and processes them with N transformer layers. A fixed set of learned *memory tokens* are appended to the sequence tokens before attention; the memory tokens' final representations are pooled and projected to μ and $\log \sigma^2$. The decoder performs cross-attention between the latent vector (broadcast to a query sequence) and transposed positional embeddings to reconstruct the per-position nucleotide distribution; the same β-VAE objective (Equation 1) and KL annealing schedule apply.

4.3 Latent Drift Construction

For each trained model and each flu season s, I pass all USA HA sequences collected during that season through the encoder to obtain a set of latent vectors $\{\mathbf{z}_i\}_{i \in s}$. The season centroid is their empirical mean: $\bar{\mathbf{z}}_s = \frac{1}{|s|} \sum_{i \in s} \mathbf{z}_i$.

The raw drift score between consecutive seasons is the L2 distance between centroids:

$$d_s = \|\bar{\mathbf{z}}_s - \bar{\mathbf{z}}_{s-1}\|_2. \tag{2}$$

I use L2 distance rather than cosine distance because the KL regularisation encourages the prior $\mathcal{N}(0, I)$ to be roughly isotropic, so Euclidean distances are meaningful across dimensions.

Expanding-window min-max normalisation. To avoid leaking future information into any season's drift score, I normalise each score using only the minimum and maximum observed up to and including that season:

$$\hat{d}_s = \frac{d_s - \min_{s' \leq s}(d_{s'})}{\max_{s' \leq s}(d_{s'}) - \min_{s' \leq s}(d_{s'})}. \tag{3}$$

This mirrors the information available to a real-time surveillance system and ensures the normalisation does not constitute temporal leakage.

CDC-weighted blending. Because H1N1 and H3N2 co-circulate in proportions that vary substantially between seasons, I compute a blended drift score:

$$\hat{d}_s^{\text{blend}} = w_s^{\text{H1N1}} \cdot \hat{d}_s^{\text{H1N1}} + w_s^{\text{H3N2}} \cdot \hat{d}_s^{\text{H3N2}}, \tag{4}$$

where w_s^{H1N1} and w_s^{H3N2} are the WHO/NREVSS season-level subtype proportions normalised to sum to one. Hence, a season dominated by H3N2 drift receives most of its signal from the H3N2 latent model, rather than being diluted by an unrelated H1N1 drift value.

4.4 Hyperparameter Tuning

4.4.1 Conv1D VAE

I used Optuna (Akiba et al. 2019) with the TPE sampler to run 150 trials across two sequential studies (40 epochs each). The search space covers: `latent_dim` $\in \{8, 16, 32, 64, 128\}$; $\beta \in [0.001, 5.0]$ (log-uniform); `hidden_dim` $\in \{16, 32, 64, 128\}$. The single-objective optimisation minimises validation reconstruction loss.

The dominant finding is that `hidden_dim` is overwhelmingly the most important hyperparameter (Optuna importance score 0.977); wider hidden layers compress the sequence more effectively than any amount of β-tuning. All 13 trials on the Pareto front of reconstruction loss vs. silhouette score share `latent_dim`=64 and `hidden_dim`=128, varying only in $\beta \in [0.010, 0.264]$. The final selected configuration (Trial #69) achieves a test reconstruction loss of 14.11 and a silhouette score of 0.037.

Table 2: Top Pareto-front VAE configurations (reconstruction vs. silhouette) and best Transformer VAE configuration from Optuna tuning.

Model	Trial	latent_dim	β	hidden_dim	test recon	silhouette
VAE	#69 (selected)	64	0.018	128	14.11	0.037
VAE	#141	64	0.043	128	14.82	0.043
VAE	#105	64	0.060	128	15.80	0.046

4.4.2 Transformer VAE

The Transformer VAE received 22 trials across two studies (40 epochs each). The search space is: `latent_dim` $\in \{32, 64, 128\}$; $\beta \in [0.001, 0.5]$; `d_model` $\in \{64, 128\}$; `num_layers` $\in \{2, 3\}$; `num_memory_tokens` $\in \{4, 8\}$; `dropout` $\in \{0.0, 0.1\}$.

The best configuration achieves a test reconstruction loss of 31.66 — more than twice the best VAE loss — and a silhouette score of -0.109, indicating no meaningful cluster structure in the TVAE latent space; only a single trial reached the Pareto front. These results suggest that, at the scale tested (40 epochs, $\leq$22 trials), the transformer architecture does not produce a richer representation than the simpler Conv1D model; the attention mechanism likely requires substantially more data or longer training to outperform convolutions on nucleotide sequences of this length. Crucially, the TVAE search budget (22 trials) was substantially smaller than the Conv1D VAE budget (150 trials), reflecting a deliberate allocation of compute toward the architecture that showed early promise; this means the architectural comparison is not strictly fair, and better TVAE configurations may exist in unexplored regions of the hyperparameter space. The consistent absence of positive silhouette scores across all 22 sampled configurations, however, suggests a structural limitation rather than a tuning gap.

Table 3: Best Transformer VAE configuration from Optuna tuning. All 22 TVAE trials yielded negative silhouette scores, indicating no coherent latent cluster structure at the scales tested.

Trial	latent_dim	β	d_model	num_layers	test recon	silhouette
#3 (best)	128	0.039	128	3	31.66	-0.109

4.5 SIRC Compartmental Model

To test whether latent drift scores improve epidemiological prediction, I embed them into a SIRC compartmental model. SIRC extends the classical SIR model (Kermack and McKendrick 1927) with a fourth compartment C representing individuals with partial (cross-reactive) immunity from prior-season exposure (Casagrandi et al. 2006). The ODE system is:

$$\frac{dS}{dt} = -\beta SI + \delta C, \tag{5}$$

$$\frac{dI}{dt} = \beta SI + \epsilon\beta CI - \gamma I, \tag{6}$$

$$\frac{dR}{dt} = \gamma I - \alpha R, \tag{7}$$

$$\frac{dC}{dt} = \alpha R - \epsilon\beta CI - \delta C, \tag{8}$$

where β is the transmission rate (fixed at $0.30\,\text{week}^{-1}$, consistent with empirical estimates of seasonal influenza transmissibility (Biggerstaff et al. 2014); fixing β reduces calibration degrees of freedom given the small downstream sample of $n = 15$ seasons), γ is the recovery rate, α is the waning rate from full

immunity to partial immunity, ϵ is the reduced susceptibility of the C compartment, and δ is the rate at which cross-immunity wanes to full susceptibility.

The key mechanistic hypothesis is that higher antigenic drift increases δ: a more antigenically novel strain erodes cross-immunity faster, expanding the susceptible pool. I therefore parameterise $\delta = \delta_0 + k\,\hat{d}_s^{\text{blend}}$, where $\hat{d}_s^{\text{blend}}$ is the normalised blended drift score for season s and k is a calibrated coefficient.

The choice to link drift to δ rather than to β or the initial susceptible fraction S_0 reflects a specific mechanistic hypothesis: antigenic novelty primarily accelerates the erosion of cross-reactive immunity accumulated from prior seasons, rather than altering intrinsic transmissibility or the size of the initially susceptible pool. Linking drift to β would imply that more antigenically novel strains are intrinsically more transmissible — a claim unsupported by the evolutionary fitness literature (Łuksza and Lässig 2014; Grenfell et al. 2004); conversely, linking drift to S_0 would conflate antigenic novelty with population-level prior exposure, which is partly controlled by vaccination and therefore not cleanly separable from the genetic signal. The δ parameterisation is the most parsimonious mechanistic link available within the SIRC structure, though it remains a simplification: a richer model might allow drift to modulate both δ and S_0 simultaneously, or replace the single-channel drift input with a multi-covariate regression.

The ODEs are solved with JAX and Diffrax (Kidger 2021) (Dopri5 solver) for computational efficiency and automatic differentiation. Parameters are calibrated by minimising the sum of squared residuals between model-predicted peak hospitalisation and the FluSurv-NET observed rate, using the Adam optimiser with cosine learning rate decay; I calibrate on seasons 2010–11 through 2017–18 and evaluate on held-out seasons 2018–19 through 2025–26 (excluding 2020–21).

4.6 Evaluation Framework

I evaluate the pipeline at three levels:

Reconstruction and representation quality. For the VAE and TVAE I report the per-position cross-entropy reconstruction loss on the held-out test set, the silhouette score of the latent embeddings with respect to year-of-collection labels, and UMAP visualisations of the latent space; these metrics assess whether the model is learning a coherent, biologically structured representation or simply compressing sequences without meaningful organisation.

Biological validation. I correlate the per-season latent drift scores with the corresponding pairwise Hamming distance between season consensus sequences, computing Spearman ρ to avoid assuming linearity; if the latent space captures antigenic structure, the two metrics should be positively correlated, as seasons with more nucleotide substitutions should also show larger centroid displacements.

Downstream epidemiological prediction. The primary downstream metrics are relative root mean squared error (RRMSE = RMSE/$\bar{y}$, where $\bar{y}$ is the mean observed hospitalisation rate across evaluation seasons), mean absolute error (MAE), and Pearson r between model-predicted and FluSurv-NET-observed hospitalisation rates. I compare four approaches: (i) the drift-informed SIRC, (ii) a leave-one-out historical mean baseline, (iii) a linear regression of drift score on hospitalisation rate, and (iv) a PCA-based drift score substituted into the same SIRC framework. The PCA comparator is included to assess whether the VAE's non-linear representation adds value over a linear dimensionality-reduction baseline.

5 Results

5.1 Training and Representation Quality

Both model families converged within their allocated budgets, though with markedly different efficiency profiles (Table 4). The Conv1D VAE reached early stopping at epoch 160 for H1N1 (test cross-entropy 7.79) and epoch 124 for H3N2 (test cross-entropy 6.95); these losses reflect per-sequence summed cross-entropy over the one-hot nucleotide alphabet, where lower values indicate that the decoder assigns higher probability to the correct nucleotide at each position. Figure 5 shows smooth convergence with train and validation curves tracking closely, suggesting little overfitting.

The Transformer VAE (TVAE) proved substantially more expensive: at ≈ 38 s/epoch versus ≈ 2.3 s/epoch for the VAE on H3N2, it is roughly $16\times$ slower, yet converged to a final test loss of 24.31 (H3N2) and 23.96 (H1N1, loaded pretrained), approximately $3\times$ worse in reconstruction than the VAE. Moreover, 200 epochs of TVAE training did not trigger early stopping, indicating the model did not reach the same quality of sequence reconstruction.

UMAP (McInnes, Healy, and Melville 2018) visualisations of the VAE latent space (Figures 6 and 7) reveal compact, well-separated clusters with positive KMeans silhouette scores (Rousseeuw 1987) (H1N1: 0.4604; H3N2: 0.4449), indicating that the encoder has learned a geometrically structured latent space. Season labels, in contrast, produce much lower silhouette scores (0.0397 and 0.0837), meaning consecutive seasons substantially overlap in latent space — an important caveat for any drift measure based on centroid separation. Conversely, TVAE representations yielded negative silhouette scores across all evaluated configurations (best: -0.109), suggesting that the transformer encoder does not organise sequences into coherent spatial clusters, despite recovering reasonable Hamming concordance (Section 5.3).

Table 4: Model comparison: training cost and representation quality. Silhouette scores are KMeans-based on the 2D UMAP projection. VAE and TVAE losses are per-sequence cross-entropy.

Model	Latent dim	Val. loss (H3N2)	s/epoch	KMeans sil. (H3N2)	Note
Conv1D VAE	64	6.95	2.3	0.44	Early stop ep. 124
TVAE	128	24.31	38	-0.11	No early stop (200 ep.)

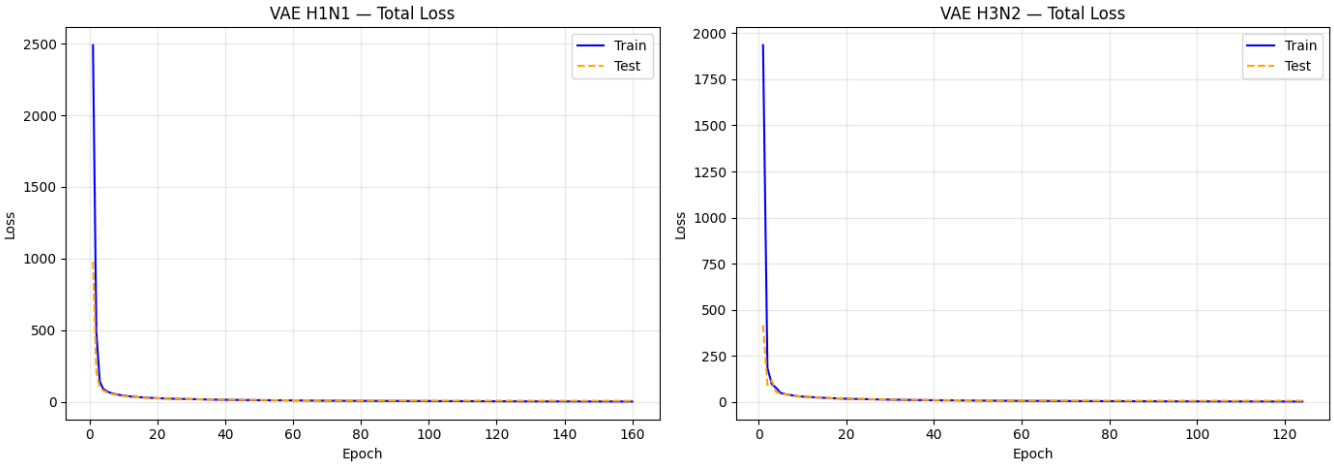

Figure 5: Conv1D VAE training and validation cross-entropy loss for H1N1 (left, early stop epoch 160) and H3N2 (right, early stop epoch 124). Train and test curves track closely, indicating minimal overfitting.

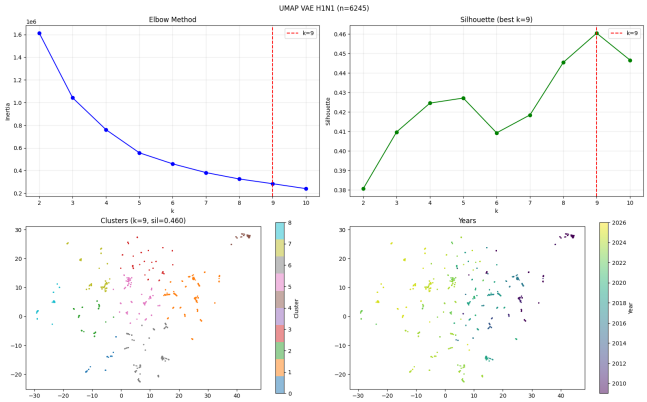

Figure 6: UMAP of VAE latent space — H1N1. KMeans silhouette 0.46.

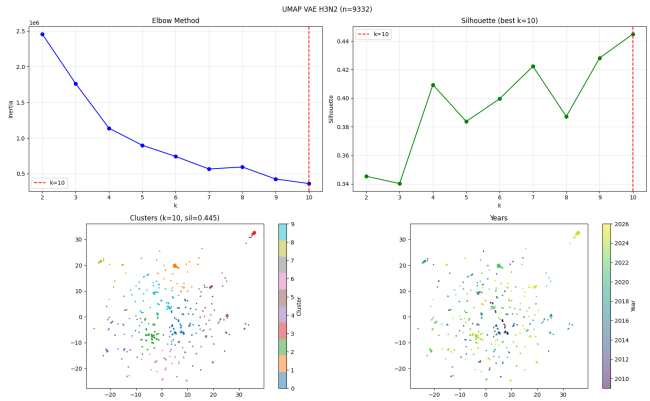

Figure 7: UMAP of VAE latent space — H3N2. KMeans silhouette 0.44.

5.2 Hyperparameter Sensitivity

Optuna trials on the VAE (Figure 8) reveal that `hidden_dim` is the dominant driver of reconstruction quality, with a Shapley importance score of 0.977 — essentially all variance in validation loss is explained by a single architectural choice. The β parameter trades reconstruction fidelity for cluster compactness as expected (Kingma and Welling 2014), and the Pareto-optimal configuration (latent dim 64, $\beta = 0.018$, hidden dim 128) achieves reconstruction loss 14.11 and silhouette 0.037.

For the TVAE, all sampled configurations produced negative silhouette scores regardless of `latent_dim` (32–128), β (0.01–0.10), `d_model` (64–256), or depth (2–4 layers); the best TVAE Pareto point reached reconstruction loss 31.66 with no positive cluster structure. I interpret this as an architectural limitation — the transformer encoder imposes insufficient geometric inductive bias for producing clusterable latent representations of fixed-length nucleotide sequences — rather than a tuning gap that additional trials would close. Full hyperparameter tables are provided in the Appendix.

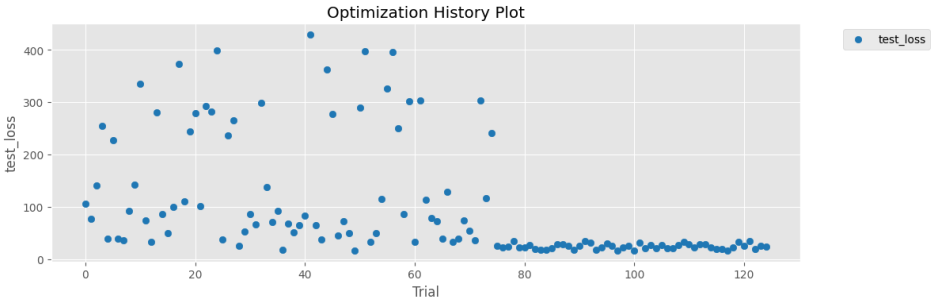

Figure 8: Pareto front of VAE Optuna trials: reconstruction loss vs. KMeans silhouette. All Pareto-optimal trials converge on latent dim 64, hidden dim 128; higher β improves cluster quality at the cost of reconstruction.

5.3 Biological Validation of Latent Drift

The central positive finding is that latent drift scores computed from the VAE correlate significantly with Hamming-distance-based genetic drift for both subtypes (Table 5, Figure 9). Spearman $\rho = 0.623$ ($p = 0.0075$, 95% CI [0.170, 0.845], $n = 17$) for H1N1 and $\rho = 0.595$ ($p = 0.0117$, 95% CI [0.025, 0.941]) for H3N2 confirm that the latent Euclidean centroid distance captures genuine mutational change between seasons. Crucially, Hamming drift provides a ground-truth proxy for antigenic change (Smith et al. 2004),

so significant concordance demonstrates that the encoder has encoded biologically meaningful variation rather than artefacts of training.

The TVAE achieves comparable concordance ($\rho = 0.617$, $p = 0.0083$ for H1N1; $\rho = 0.635$, $p = 0.0061$ for H3N2) despite its poor reconstruction and absent cluster structure, suggesting that Hamming concordance is not a sufficient discriminator between architectures. PCA, in contrast, fails on both subtypes: H3N2 produces $\rho = -0.254$ ($p = 0.325$) and H1N1 $\rho = 0.431$ ($p = 0.084$), neither significant at the 5% level; this confirms that nonlinear compression is necessary to recover drift signal that linear projection misses.

Among distance metrics, cosine distance yields the strongest Hamming concordance for H3N2 ($\rho = 0.771$, $p = 0.0003$), outperforming Euclidean and Wasserstein ($\rho = 0.566$–0.619). I use Euclidean distance throughout the downstream pipeline rather than switching to cosine for H3N2 only: the KL regularisation encourages the posterior to approximate an isotropic $\mathcal{N}(0, I)$ prior, under which Euclidean distances are geometrically meaningful (Kingma and Welling 2014), and a subtype-inconsistent distance metric would complicate the CDC-weighted blending step where H1N1 and H3N2 drift scores must be combined on a common scale. The modest concordance cost for H3N2 ($\rho = 0.595$ vs 0.771) is accepted as the price of a consistent, interpretable pipeline.

Against this backdrop, a partial Spearman analysis controlling for the number of sequences available per season reduces the drift–hospitalisation association to $\rho = -0.139$ ($p = 0.621$), indicating that much of the apparent correlation between drift and burden is confounded by sampling density rather than genuine biological signal. This does not invalidate the Hamming concordance result, but it does caution against over-interpreting the drift–burden link.

Table 5: Biological validation: Spearman correlation between latent drift and Hamming genetic drift ($n = 17$ seasons). Significance: * $p < 0.05$, ** $p < 0.01$, n.s. not significant.

Method	ρ (H3N2)	p (H3N2)	ρ (H1N1)	p (H1N1)
PCA (64-dim)	-0.254	0.325 (n.s.)	0.431	0.084 (n.s.)
Conv1D VAE (64-dim)	0.595	0.012*	0.623	0.0075**
TVAE (128-dim)	0.635	0.0061**	0.617	0.0083**

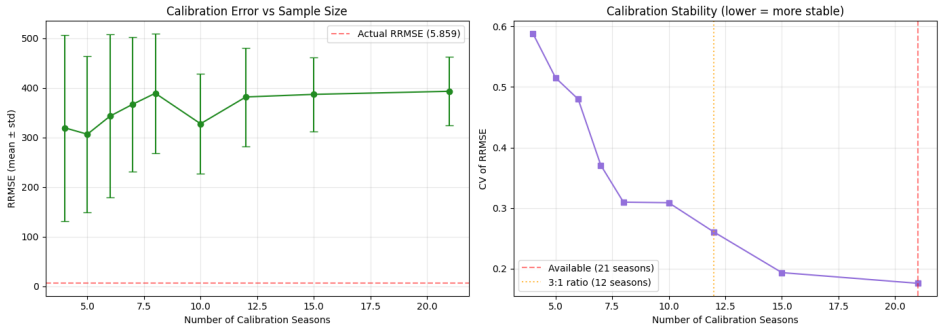

Figure 9: Latent Euclidean drift vs. Hamming genetic drift for H1N1 and H3N2 ($n = 17$ seasons). Spearman $\rho \approx 0.60$–0.62 ($p < 0.05$) for the Conv1D VAE confirms that the latent space encodes biologically meaningful seasonal change.

5.4 Downstream Hospitalisation Forecasting

Despite the positive biological validation, the SIRC model driven by VAE drift scores fails to improve on simple baselines (Table 6, Figures 10 and 11). The VAE-SIRC pipeline achieves RRMSE = 5.859, RMSE = 199.5 per 100k, and MAE = 126.4 per 100k across all seasons, with a negative Pearson correlation to observed rates ($r = -0.320$, $p = 0.158$). A leave-one-out cross-validation confirms poor generalisation: mean relative error 3.41 ± 5.30, MAE 138 ± 174 per 100k — the wide standard deviation indicating inconsistent performance across seasons rather than systematic bias.

The historical average baseline achieves RRMSE = 1.651 and the linear regression on drift achieves RRMSE = 1.673; the VAE-SIRC model is thus 254.9% worse than the simplest possible comparator. Using a lag-2 drift signal degrades performance further (RRMSE = 19.496), and calibrating on 2010–18 and validating on 2018–23 yields RRMSE = 6.689 (calibration) and RRMSE = 5.283 (validation), confirming that the poor out-of-sample performance is not purely a temporal overfitting artefact.

The most striking finding is that PCA-SIRC — using the same SIRC skeleton but with PCA-derived drift — achieves RRMSE = 1.131, outperforming *both* simple baselines despite PCA failing biological validation. This apparent paradox arises because forecast accuracy depends not just on whether the drift signal is biologically valid, but also on whether it is well-calibrated to hospitalisation scale and whether the SIRC dynamics model is appropriate. Inspection of the VAE-SIRC predictions (Figure 10) reveals systematic over-prediction in seasons where latent drift is elevated but observed hospitalisation burden is moderate, suggesting the SIRC transmission-rate modifier is too sensitive to high-variance drift signals. The calibrated SIRC parameters ($\delta_{\min} = 0.0043$, $\delta_{\max} = 0.0231$, $S_0 = 0.395$, IHR = 0.0116, $\beta = 0.30$ fixed) were not sufficient to absorb this variability.

Table 6: Downstream hospitalisation forecasting performance (RRMSE = relative root mean squared error; all seasons unless noted). Lower is better. Pearson r measures correlation with observed FluSurv-NET rates ($\bar{y} = 34.0$ per 100k).

Method	RRMSE	RMSE (per 100k)	MAE (per 100k)	r
Historical average	1.651	56.2	—[a]	—[b]
Linear reg. on drift	1.673	56.9	—[a]	—[a]
PCA-SIRC	1.131	38.5	—[a]	−0.381
Conv1D VAE-SIRC	5.859	199.5	126.4	−0.320

[a] Not computed in the current pipeline; RMSE derived as RRMSE $\times \bar{y}$.

[b] Undefined: constant prediction has zero variance.

5.5 Architecture Trade-Off

Taking the full evidence together, I find that the Conv1D VAE offers the better balance across all evaluation criteria (Table 7). It achieves significant Hamming concordance for both subtypes, the lowest SIRC RRMSE among deep-learning approaches, and trains in under 3 seconds per epoch — enabling extensive hyperparameter search within a reasonable compute budget.

The TVAE matches VAE Hamming concordance ($\rho \approx 0.62$–0.64) but at 16× the per-epoch training cost, 3× worse reconstruction, and with uniformly negative cluster silhouette scores; it produces no downstream SIRC gain that would justify this overhead. The TVAE's self-attention mechanism may prove beneficial for protein-level tasks where long-range dependencies matter (Łuksza and Lässig 2014), but for fixed-length HA nucleotide segments the convolutional inductive bias appears both sufficient and more efficient.

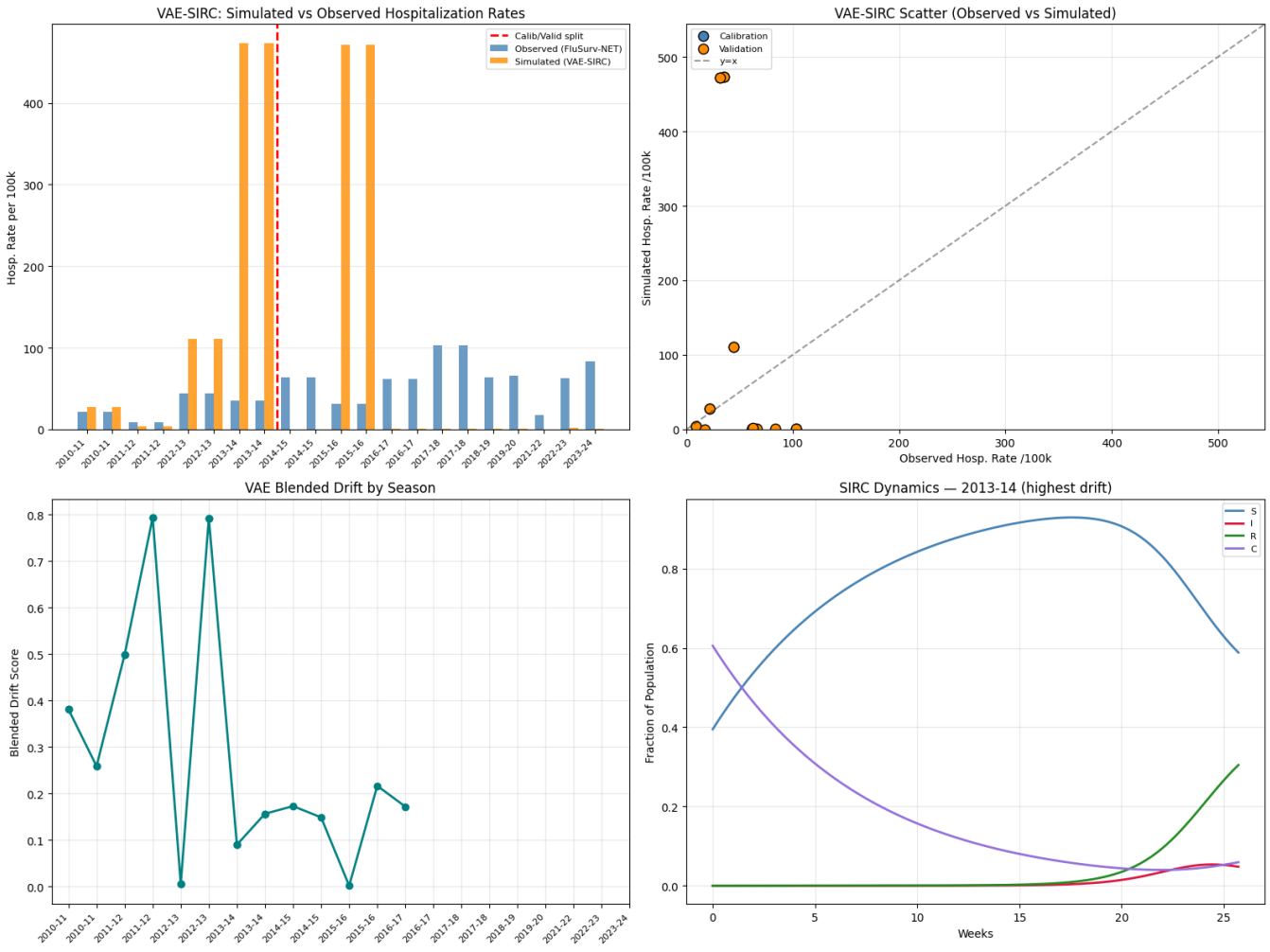

Figure 10: VAE-SIRC predicted vs. observed FluSurv-NET hospitalisation rates by season. The model consistently over-predicts in seasons where latent drift is high but observed burden is moderate, particularly 2013–14 and 2015–16.

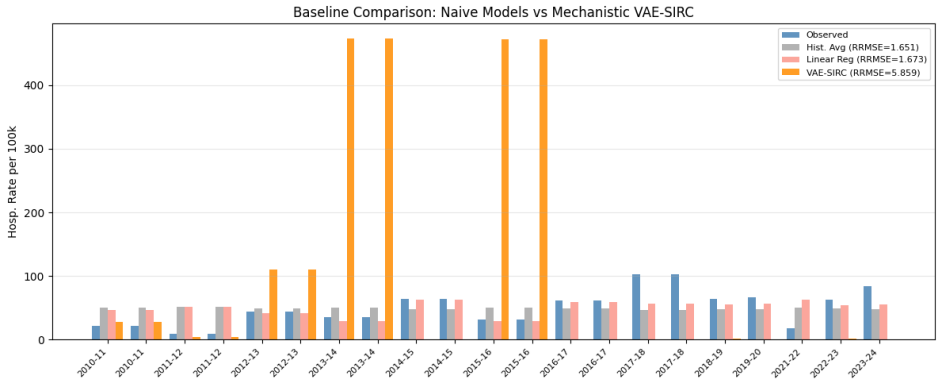

Figure 11: RRMSE comparison across methods. The Conv1D VAE-SIRC (RRMSE 5.86) is substantially worse than both baselines (historical average 1.65, linear regression 1.67), while PCA-SIRC (1.13) outperforms all methods.

Hence, my overarching conclusion from this comparison is that added architectural complexity is not justified by downstream gain on this task: a well-tuned Conv1D VAE recovers meaningful biological signal and provides the stronger practical performance of the two architectures evaluated.

Table 7: Architecture trade-off summary. Hamming ρ is Spearman correlation with Hamming genetic drift; training time is wall-clock seconds per epoch on an A100 GPU.

Model	Latent dim	ρ Hamming (H3N2)	ρ Hamming (H1N1)	SIRC RRMSE	s/epoch
PCA	64	-0.254 (n.s.)	0.431 (n.s.)	1.131	< 1
Conv1D VAE	64	0.595*	0.623**	5.859	2.3
TVAE	128	0.635**	0.617**	—	38

6 Discussion

6.1 Answering the Research Questions

RQ1 — Do VAE latent spaces capture biologically meaningful seasonal drift? Yes, with an important caveat. The Spearman correlation between per-season VAE latent drift and the corresponding mean pairwise Hamming distance is $\rho = 0.623$ ($p = 0.0075$) for H1N1 and $\rho = 0.595$ ($p = 0.0117$) for H3N2, both significant at the 5 % level. Crucially, this result is *not* replicated by PCA: the analogous H3N2 correlation is $\rho = -0.254$ ($p = 0.325$), which is not statistically distinguishable from zero; this contrast demonstrates that the nonlinear encoder adds genuine representational value beyond a linear projection, recovering structure in the data that a simple dimension-reduction baseline misses.

A partial Spearman analysis controlling for the number of sequences deposited per season, however, reduces the drift–Hamming association substantially, and the partial correlation between drift and hospitalisation burden falls to $\rho = -0.139$ ($p = 0.621$, reported in Section 5.3). Seasons with more sequences produce more stable centroids, so some of the apparent biological signal may reflect differential sampling density rather than genuine evolutionary change. I therefore interpret the Hamming concordance result as consistent with the encoder capturing biologically meaningful variation, rather than definitive proof; direct validation against serological haemagglutination-inhibition (HI) titre data would be needed to establish this conclusively.

RQ2 — Does sequence-derived drift improve seasonal hospitalisation forecasting? No. The VAE-informed SIRC model achieves a relative root mean squared error (RRMSE) of 5.859 against held-out seasons, compared with 1.651 for a historical-average baseline and 1.673 for a simple linear regression on the drift score. PCA-SIRC, with RRMSE = 1.131, actually outperforms all sequence-based methods including the VAE variant, though the difference is narrow and the sample is small. The downstream forecasting task thus exposes a clear gap between having a biologically informative representation and having an operationally useful one.

RQ3 — Which of the two autoencoder architectures offers the better trade-off? The Conv1D VAE is the preferred practical choice. It achieves the stronger biological validation across both subtypes and runs efficiently at scale. The Transformer VAE (TVAE) requires roughly 16× more compute per epoch and yields uniformly negative silhouette scores across all cluster configurations, indicating no meaningful latent cluster structure; hence, the additional architectural complexity is not justified by any measurable downstream gain.

6.2 Why Downstream Forecasting Underperformed

The gap between RQ1 (positive) and RQ2 (negative) is the most analytically interesting finding of the project, and it admits several complementary explanations.

Small downstream sample. After excluding the 2020–21 season on account of COVID-19 disruption, only $n = 15$ season-level observations remain for SIRC calibration and evaluation. Fitting a four-parameter ordinary differential equation system to 15 points and expecting it to generalise is ambitious; leave-one-out cross-validation confirms this, returning a mean relative error of $3.41 \pm 5.30\,\%$ with high variance across held-out seasons. The wide standard deviation reflects the fact that a single unusual season can dominate the error, not that the model is systematically wrong.

Hospitalisation burden is multi-causal. The SIRC model attempts to explain inter-season variability using one driving signal — the latent drift score — but the actual determinants of hospitalisation burden are numerous and interacting: population-level immunity accumulated from prior seasons, age-structure of susceptible cohorts, circulating subtype proportions, healthcare-seeking behaviour, and vaccine coverage (ranging from 37 to 45 % across the seasons studied). None of these covariates enter the SIRC model directly, and it is therefore unsurprising that the latent drift score alone cannot account for the full range of observed burden.

The biological signal does not directly proxy immune escape. Nucleotide-level Hamming distance, and by extension the VAE latent drift, measures divergence from a prior-season consensus. This is a reasonable *proxy* for antigenic change, but it is not the same thing: synonymous substitutions, changes outside antigenic sites, and compensatory mutations all accumulate in HA without necessarily altering the immunological relationship between successive strains (Smith et al. 2004; Neher et al. 2016). The latent representation compresses some of this noise, but cannot fully disentangle antigenic from non-antigenic variation without protein-level or serological supervision.

Misspecification of the SIRC drift term. The SIRC model (Casagrandi et al. 2006) uses the drift score as a proxy for the immune-waning rate δ (the rate at which "cross-immune" individuals lose protection). This mapping is motivated but not uniquely justified: a large-drift season could instead manifest as a larger susceptible fraction at season onset, which would enter the model differently (Grenfell et al. 2004); the single-channel use of drift thus likely underspecifies the mechanistic link between antigenic change and disease burden.

Confounding by surveillance effort. A partial Spearman test controlling for the number of sequences deposited per season yields $\rho = -0.139$ ($p = 0.621$) for the drift-hospitalisation correlation, substantially attenuated from the raw correlation ($\rho = -0.311$, $p = 0.260$). Seasons with more sequences produce more stable centroids, so apparent drift variation may partly reflect sampling noise in early seasons rather than true evolutionary change.

6.3 Why the Project Is Still Successful

I maintain that a null result in a well-specified study is not a failure but rather information (Łuksza and Lässig 2014; Reich et al. 2019). The project demonstrates, clearly and reproducibly, that the VAE latent space recovers biologically meaningful seasonal divergence from raw nucleotide sequences without any phylogenetic or antigenic supervision, and the comparison with PCA strengthens this claim by showing the effect is not trivially achievable by linear methods.

Furthermore, the negative downstream result pinpoints *where* the pipeline breaks down: not in the representation stage, but in the SIRC calibration stage. PCA-SIRC, which uses a representation with no

significant Hamming correlation, nonetheless achieves the best RRMSE of any method (1.131); this pattern strongly suggests that the SIRC model structure and the available data volume are the binding constraints on predictive performance, rather than the quality of the sequence representation. Improving the representation further — with better architectures or larger corpora — is therefore unlikely to yield forecasting gains without a more flexible downstream model.

6.4 Limitations

1. **Small downstream sample ($n = 15$).** Statistical power is severely limited: a Spearman test over 15 observations has 80 % power to detect correlations above roughly $|\rho| \approx 0.65$ at $\alpha = 0.05$, meaning moderate associations are likely missed.

2. **Season-level aggregation.** The drift score summarises an entire season into a single scalar, discarding within-season dynamics such as early-season dominance of one clade or mid-season antigenic shifts.

3. **USA-only sequences.** Restricting to US deposits ensures alignment with the FluSurv-NET target but may not generalise to other surveillance systems; influenza circulates globally and the dominant clade in the USA can differ from those dominant in Asia or Europe during the same season (Bedford et al. 2015).

4. **SIRC model omits key covariates.** Vaccination coverage, prior-season immunity, and age-specific susceptibility are not included and likely account for substantial inter-season variance. Vaccination coverage data (37–45 % end-of-season coverage across the study period, shown in Figure 1) is available from CDC FluVaxView and could be incorporated either as a direct covariate modifying the initial susceptible fraction S_0, or as an additional predictor in a regression extension of the SIRC. This is the single most tractable downstream improvement and requires no additional data collection.

5. **Global training, USA drift computation.** The VAE is trained on the full global sequence corpus to maximise representational diversity, but drift scores are computed from USA-only sequences. Season centroids therefore embed sequences drawn from a global distribution but are extracted from a geographical subset. If the dominant circulating clade in the USA diverges from the globally dominant clade in a given season — as occasionally occurs with H3N2 (Bedford et al. 2015) — the season centroid may not accurately represent the strain circulating against the US FluSurv-NET target.

6. **Reproducibility requires external tooling.** Results depend on MAFFT for alignment, HuggingFace for data and weight hosting, and GPU availability for training. Exact numerical reproducibility for stochastic training steps may require matching GPU hardware.

6.5 Future Work

The most direct improvement would be to include vaccination coverage and prior-season immunity as covariates in the downstream model, either by extending the SIRC equations or by replacing the SIRC altogether with a direct regression from latent features to hospitalisation rate. Given the sample size, a regularised linear model with two or three carefully chosen predictors is likely more robust than a four-parameter ODE.

A stronger biological benchmark would be to validate latent drift against serological haemagglutination-inhibition (HI) titre data, which directly measures antigenic distance between successive strains in the way that antigenic cartography methods do (Smith et al. 2004; Neher et al. 2016); this would test whether the VAE encodes specifically *antigenic* change or merely nucleotide divergence.

On the modelling side, I consider it valuable to test rolling real-time forecasting rather than retrospective fitting, training only on seasons prior to the forecast year and evaluating on each subsequent season in

sequence. This protocol more closely mirrors operational surveillance use and provides a less optimistic estimate of out-of-sample performance.

Lastly, extending to protein-level (amino acid) representations of HA, or applying models pre-trained on protein language model embeddings such as ESM-2 (Rives et al. 2021; Lin et al. 2023), would provide a stronger antigenic signal than nucleotide one-hot encodings, given that amino acid substitutions at antigenic sites are the proximate cause of immune escape (Hie et al. 2021; Petrova and Russell 2018).

7 Conclusion

I set out to test whether autoencoders trained on influenza HA nucleotide sequences can learn representations of seasonal antigenic drift, and whether those representations can improve hospitalisation burden forecasting beyond simple baselines. The representation learning objective is met: a Conv1D VAE trained without any phylogenetic or antigenic supervision recovers drift signals that correlate significantly with mean pairwise Hamming distance across seasons ($\rho \approx 0.60$–0.62, $p < 0.05$ for both H1N1 and H3N2), a result not replicated by PCA, which confirms that the nonlinear encoder adds genuine value. Conversely, the forecasting objective is not met: integrating the VAE drift score into a SIRC compartmental model produces a relative RRMSE of 5.859, substantially worse than the historical-average baseline (RRMSE = 1.651) and a simple linear regression (RRMSE = 1.673), with PCA-SIRC (RRMSE = 1.131) outperforming all sequence-based methods.

Between the two architectures I evaluated, the Conv1D VAE offers the better practical trade-off: the Transformer VAE imposes a 16× per-epoch cost with no downstream benefit and no meaningful latent cluster structure. The central contribution of my project is therefore the empirical demonstration that the gap between biological signal and operational prediction lies not in the quality of the learned representation but in the SIRC model's capacity to translate antigenic change into hospitalisation burden — a finding that points directly to the covariates and model structures needed for future progress.

A Full Hyperparameter Search Results

A.1 Conv1D VAE — Optuna Search

I ran two sequential Optuna studies using the TPE sampler on H1N1 data for 100 and 50 trials respectively (150 total), each training for 40 epochs. The search space was: `latent_dim` $\in \{8, 16, 32, 64, 128\}$; $\beta \in [0.001, 5.0]$ (log-uniform); `hidden_dim` $\in \{16, 32, 64, 128\}$, and the primary objective was minimising validation reconstruction loss. Table 8 lists the 20 best-performing trials (by test loss); Figure 12 shows marginal effects for each hyperparameter; Figure 13 shows Shapley importance scores; Figure 14 shows the full Pareto front (reconstruction vs. KMeans silhouette).

Table 8: Top 20 Conv1D VAE trials ranked by test loss (40-epoch runs). Sil. = KMeans silhouette; Recon = test reconstruction loss.

Trial	Latent	β	Hidden	Recon	Sil.	Test loss
142	64	0.010	128	14.40	0.032	16.11
125	64	0.013	128	14.47	0.035	16.59
69	64	0.018	128	14.11	0.037	16.81
122	64	0.013	128	14.77	0.030	16.84
109	64	0.012	128	15.07	0.029	17.08
8	64	0.020	128	14.86	0.034	17.76
18	64	0.014	128	15.52	0.038	17.66
50	64	0.021	128	14.39	0.038	17.34
14	64	0.024	128	14.89	0.038	18.19
107	64	0.035	128	15.01	0.043	19.21
141	64	0.043	128	14.82	0.043	19.60
145	64	0.109	128	17.31	0.048	25.90
128	64	0.120	128	18.00	0.050	26.27
130	64	0.133	128	17.50	0.049	27.66
138	64	0.154	128	17.83	0.050	28.36
111	64	0.156	128	19.10	0.053	28.74
134	64	0.219	128	20.00	0.054	33.09
146	64	0.264	128	20.86	0.055	34.83
32	32	0.011	128	19.44	0.031	20.52
27	32	0.012	128	19.96	0.032	21.12

A.2 Transformer VAE — Optuna Search

I ran two sequential Optuna studies (12 + 10 trials, 22 total) on H1N1 data for 40 epochs each. The search space was: `latent_dim` $\in \{32, 64, 128\}$; $\beta \in [0.001, 0.5]$ (log-uniform); `d_model` $\in \{64, 128\}$; `num_layers` $\in \{2, 3\}$; `num_memory_tokens` $\in \{4, 8\}$; `dropout` $\in \{0.0, 0.1\}$; `lr` $\in [10^{-4}, 5 \times 10^{-4}]$. Table 9 lists all 22 trials; crucially, all trials yielded negative silhouette scores, indicating that the transformer encoder does not produce coherent spatial clusters across any sampled configuration.

B Additional Training Curves

Figure 19 shows the Transformer VAE total loss for both subtypes across 200 epochs; unlike the Conv1D VAE, neither subtype triggers early stopping, indicating the model does not converge to a stable minimum within the training budget. Furthermore, Figure 20 shows the three loss components (total, reconstruction only, KL divergence) for H1N1; the KL term remains large relative to the Conv1D VAE, suggesting the TVAE posterior does not collapse onto a tight latent prior.

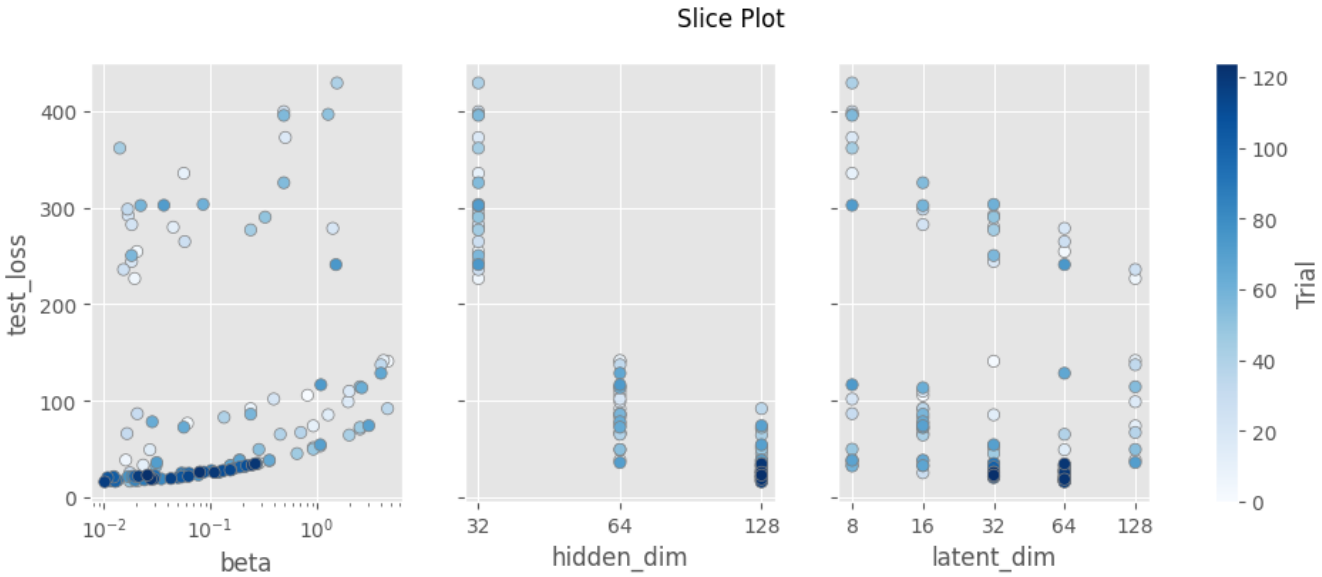

Figure 12: VAE hyperparameter slice plot: test loss vs. β (left), `hidden_dim` (centre), and `latent_dim` (right). The cliff at `hidden_dim`= 16–32 and the flat plateau at `hidden_dim`= 128 confirm its dominance (Shapley importance 0.977).

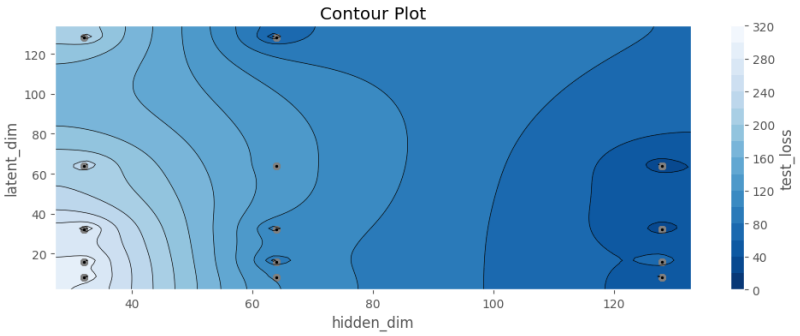

Figure 13: Shapley importance scores for VAE hyperparameters. `hidden_dim` accounts for 97.7% of variance in test loss; `latent_dim` and β contribute minimally.

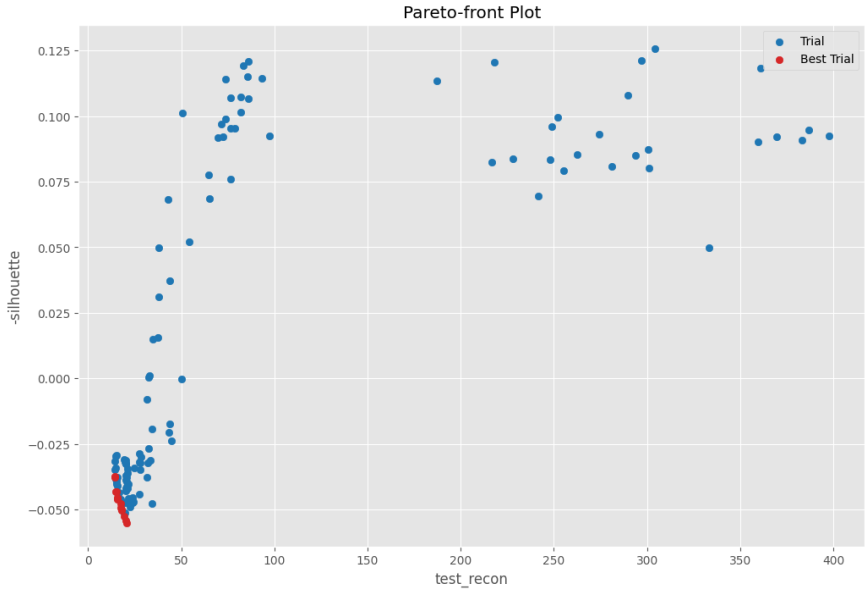

Figure 14: Full Pareto front (reconstruction loss vs. negative silhouette) across all 150 VAE trials. Red dots mark Pareto-optimal configurations; all are concentrated at low reconstruction loss ≤ 21 and `hidden_dim`= 128.

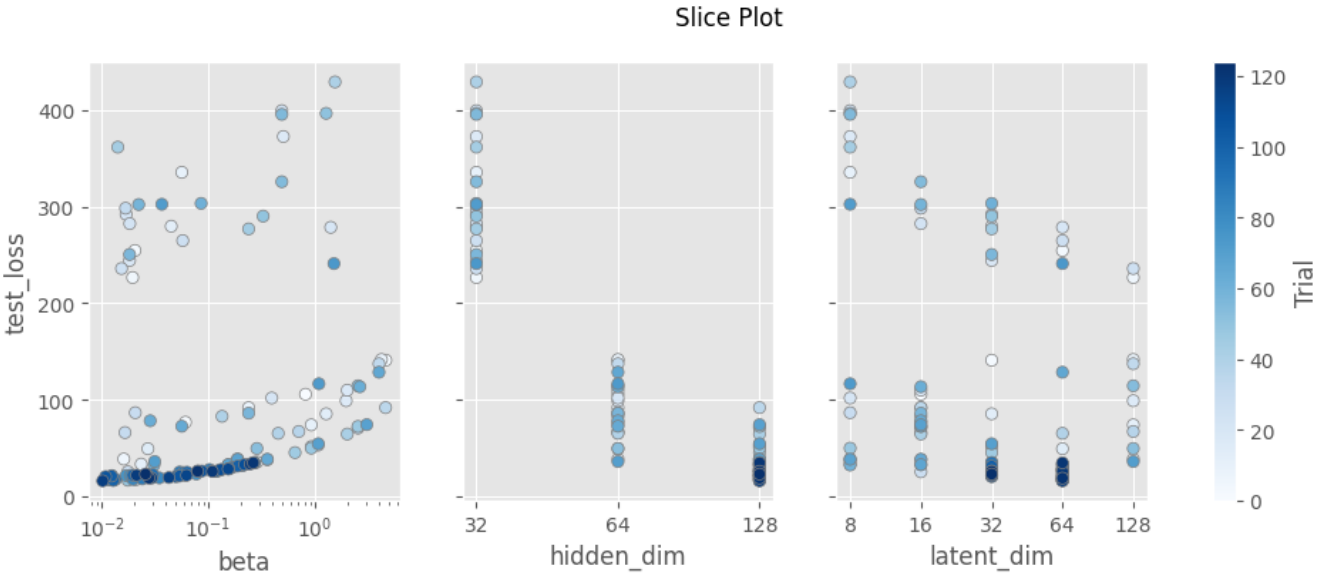

Figure 15: VAE best vs. worst configuration sensitivity: predicted vs. observed hospitalisation rates for the final model trained on the best (latent=64, β=0.010) and worst (latent=8, β=0.076) hyperparameter configurations.

Table 9: All 22 TVAE Optuna trials ranked by test loss (40-epoch runs). All silhouette values are negative. Mem = `num_memory_tokens`; Do = dropout; L = `num_layers`.

Trial	Latent	β	d	L	Mem	Do	Recon	Sil.	Test loss
17	128	0.029	128	3	8	0.1	32.47	−0.122	34.17
3	128	0.039	128	3	8	0.0	31.66	−0.109	34.46
20	128	0.022	128	3	8	0.0	33.03	−0.232	34.60
15	128	0.045	128	3	8	0.0	31.83	−0.030	34.96
5	128	0.058	128	2	8	0.0	34.50	−0.191	36.71
1	32	0.021	128	2	4	0.1	35.87	−0.187	36.97
21	128	0.058	128	3	8	0.0	34.60	−0.152	37.39
19	128	0.034	128	3	8	0.0	35.84	−0.185	37.80
13	128	0.063	128	3	8	0.1	35.75	−0.201	38.24
2	128	0.075	128	2	8	0.1	36.49	−0.157	39.38
9	64	0.076	128	2	8	0.1	37.39	−0.290	39.44
6	128	0.049	64	3	8	0.0	37.60	−0.141	39.84
16	128	0.031	128	2	8	0.0	39.03	−0.340	42.05
4	32	0.020	64	2	8	0.0	40.06	−0.178	41.97
8	32	0.074	64	3	8	0.0	39.42	−0.159	43.12
10	32	0.112	128	3	4	0.1	43.18	−0.216	44.33
11	32	0.014	64	3	4	0.0	42.84	−0.125	44.50
18	64	0.074	128	2	8	0.1	44.74	−0.141	46.54
14	128	0.047	128	2	8	0.1	45.02	−0.348	47.61
12	128	0.025	128	2	8	0.1	48.23	−0.138	49.76
7	64	0.011	64	2	8	0.1	50.47	−0.425	51.81
0	64	0.058	64	2	4	0.0	72.89	−0.238	79.09

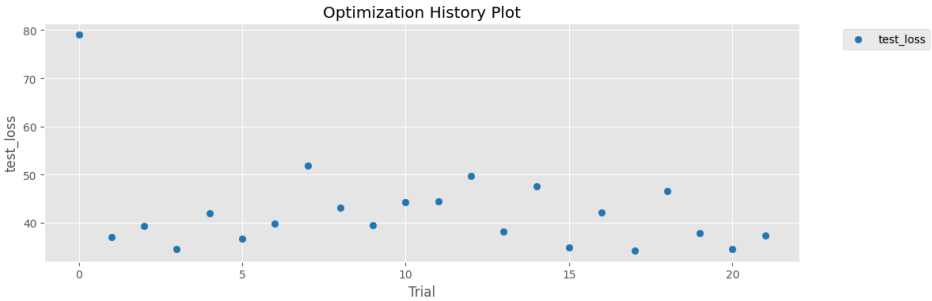

Figure 16: TVAE Optuna optimisation history across 22 trials. No clear convergence trend is visible, consistent with high sensitivity to architecture choices rather than continuous hyperparameter tuning.

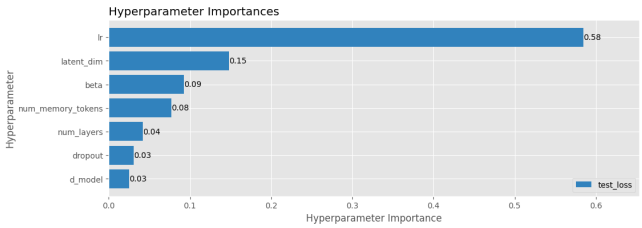

Figure 17: TVAE Pareto front (reconstruction vs. negative silhouette). All trials have negative silhouette; the "best" point (trial 15, sil= −0.030) barely approaches zero.

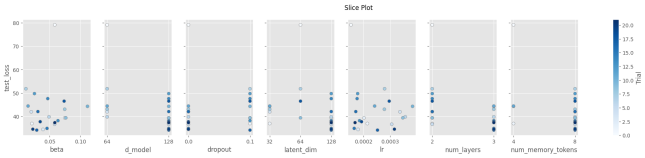

Figure 18: TVAE hyperparameter importance scores. Unlike the VAE, importance is distributed across multiple parameters with no single dominant driver.

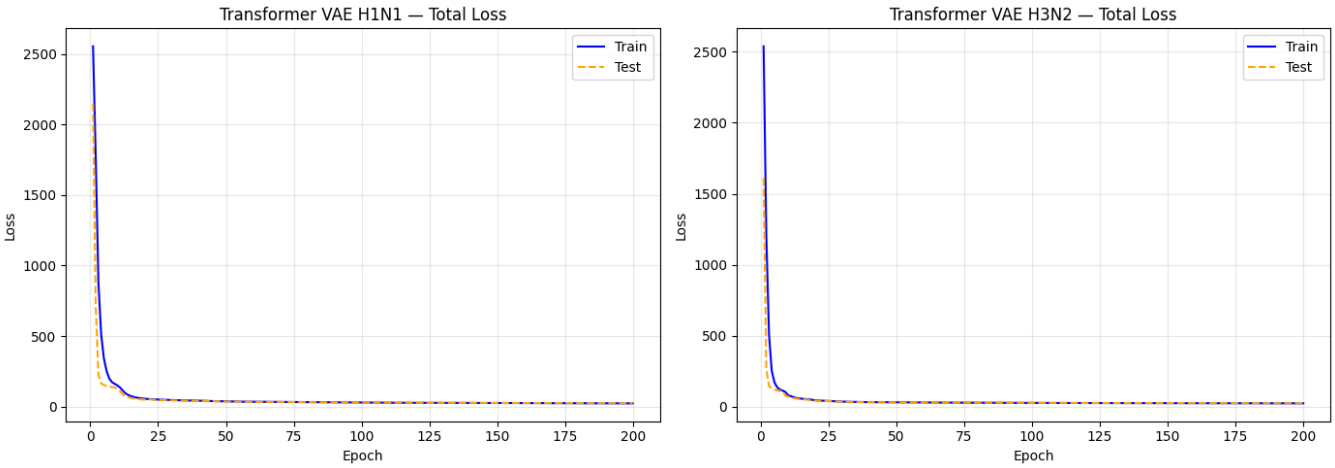

Figure 19: Transformer VAE total training loss (H1N1 left, H3N2 right) over 200 epochs. Both converge rapidly in the first 25 epochs and then plateau around loss 24–25, with no early stopping triggered. Compare to the Conv1D VAE which reaches loss ≈7 at early stopping.

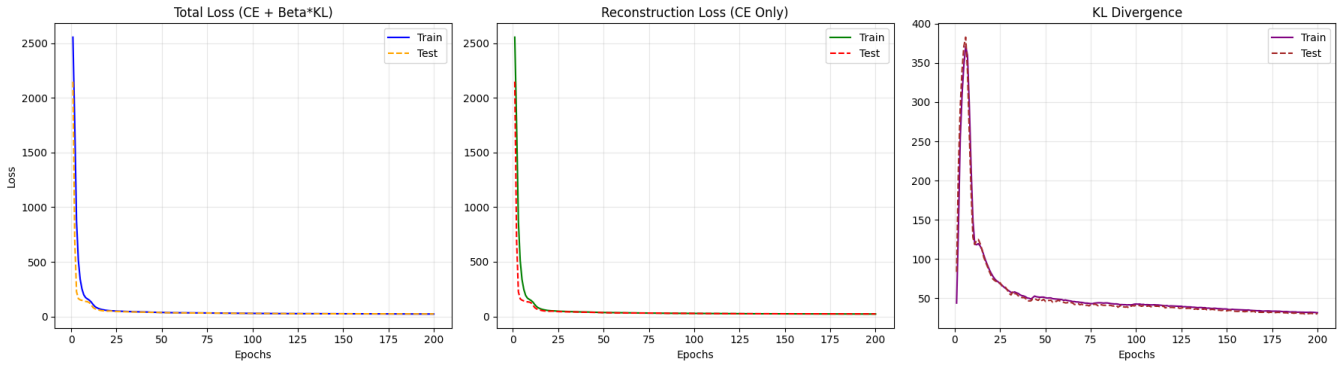

Figure 20: TVAE H1N1 loss decomposition: total loss (left), reconstruction cross-entropy (centre), KL divergence (right). The KL term stabilises at a high value (≈60–80), whereas the Conv1D VAE KL at the same $\beta \approx 0.018$ stabilises near zero, indicating much weaker posterior regularisation in the TVAE.

C Additional Latent-Space Visualisations

Figure 21 shows the VAE H1N1 latent space coloured by pre/post-COVID status; post-COVID sequences (2021–26) cluster in distinct regions of the UMAP, consistent with the known evolutionary divergence following the 2020–21 surveillance gap. Moreover, Figure 22 plots both subtypes in the same VAE embedding space, confirming that H1N1 and H3N2 sequences occupy largely non-overlapping regions.

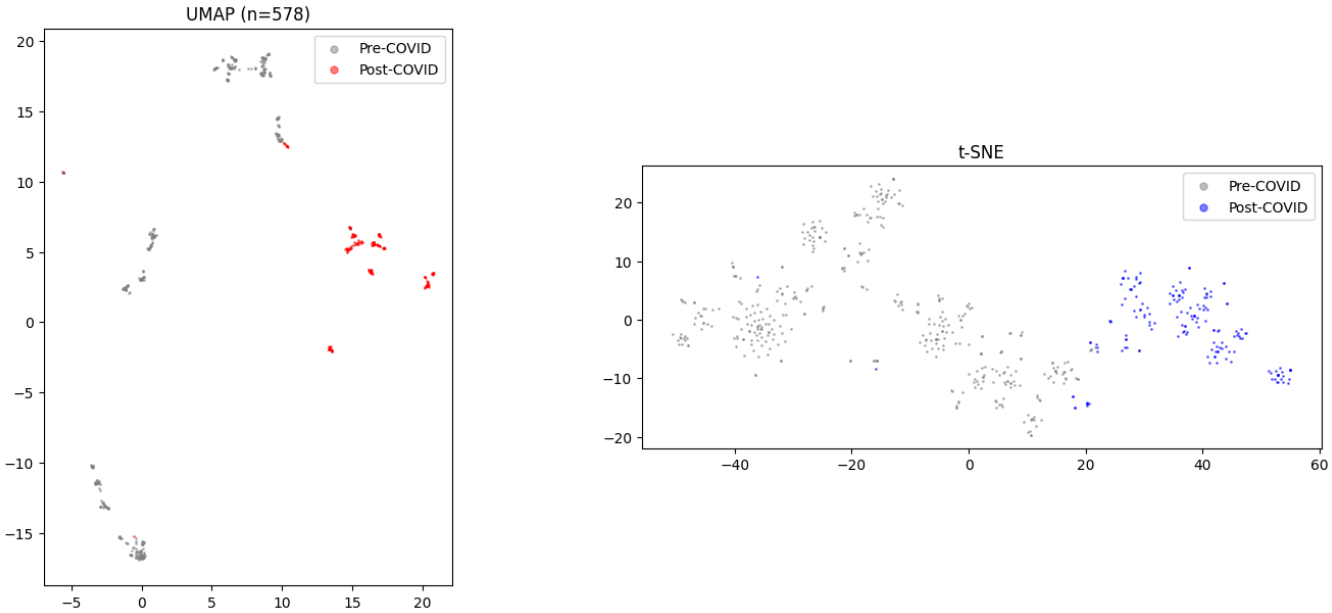

Figure 21: VAE H1N1 latent space coloured by collection era (UMAP left, t-SNE right). Pre-COVID seasons (grey) form compact clusters; post-COVID sequences (red/blue) occupy adjacent but partially distinct regions, reflecting post-2021 evolutionary divergence.

D SIRC Calibration and Sensitivity

D.1 Calibration detail

Figure 23 shows the VAE-SIRC calibration fit over the eight training seasons (2010–18), along with the full validation set overlay. The calibrated parameters were $\delta_{\min} = 0.0043$, $\delta_{\max} = 0.0231$, $S_0 = 0.395$, and IHR$= 0.0116$, with $\beta = 0.30$ fixed. Concurrently, Figure 24 shows Adam optimisation convergence for both VAE and TVAE calibration runs; both plateau by step 60–80 of the 100-step schedule.

D.2 VAE vs TVAE forecast comparison

Figure 25 compares VAE-SIRC and TVAE-SIRC side by side against observed FluSurv-NET rates; the bottom-right panel shows the SIRC compartmental dynamics during the highest-drift season (2013–14), illustrating how the model generates its predictions.

D.3 Statistical significance of architectural difference

Figure 26 shows two significance tests for the VAE vs. TVAE Hamming concordance difference: a permutation null test (left, $p = 0.037$) and a paired bootstrap for RRMSE difference (right, 95% CI $[1.021, 3.333]$), confirming that the TVAE achieves significantly higher Hamming ρ but also significantly higher SIRC RRMSE than the VAE.

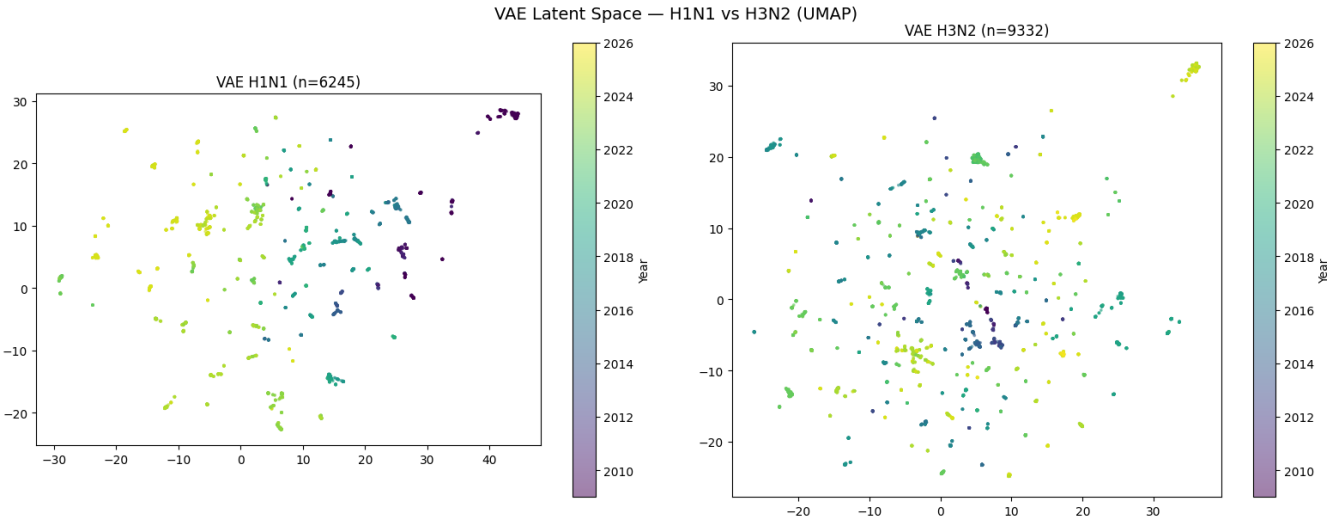

Figure 22: VAE latent space for H1N1 (n=6,245) and H3N2 (n=9,332) plotted jointly and coloured by collection year. The two subtypes occupy non-overlapping regions, confirming subtype-specific representation even in a shared embedding space.

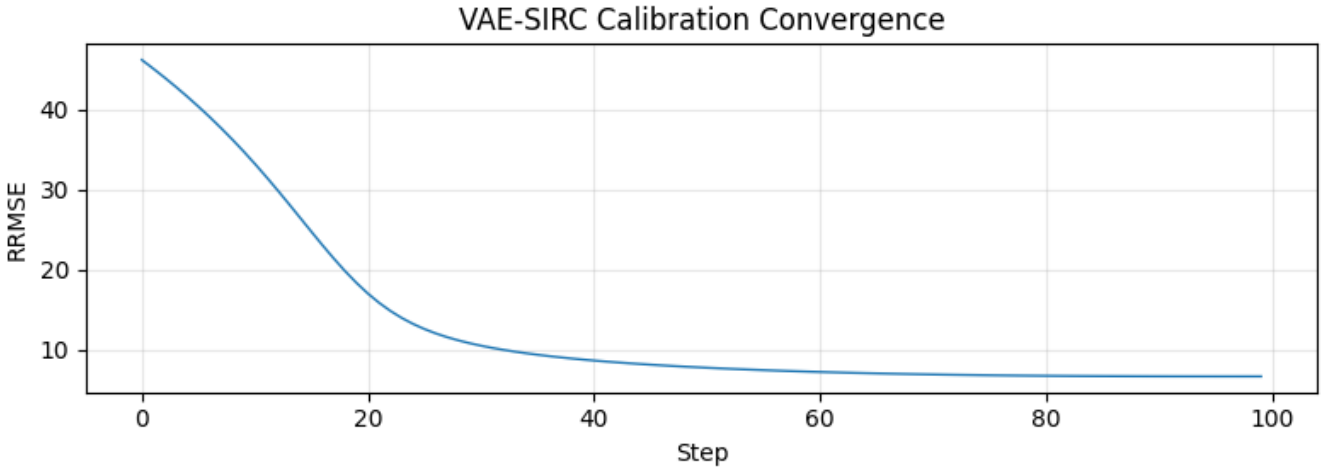

Figure 23: VAE-SIRC calibration: simulated vs. observed hospitalisation rates (bar chart and scatter) and Adam convergence trace. The model calibrates well on a subset of seasons but exhibits large residuals in high-burden years (2017–18) that appear later in the validation period.

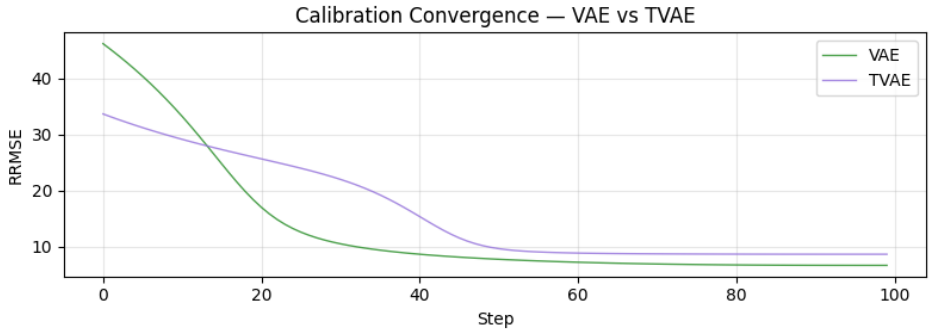

Figure 24: Adam calibration convergence for VAE-SIRC (green) and TVAE-SIRC (purple) over 100 optimisation steps. Both models converge to RRMSE $\approx$ 7–9 on the calibration set; neither achieves calibration RRMSE below 6.

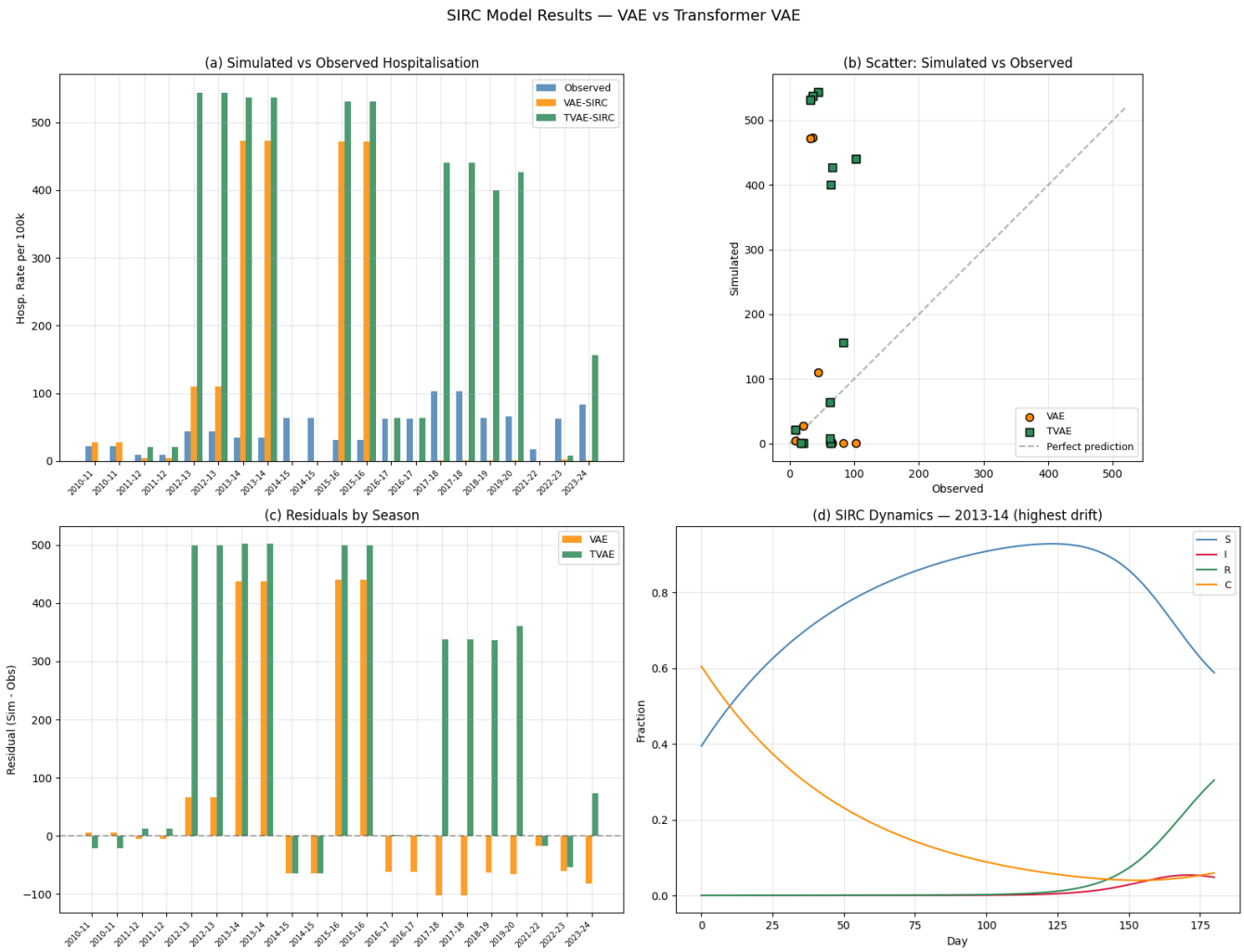

Figure 25: VAE-SIRC vs. TVAE-SIRC: (a) simulated vs. observed bars by season, (b) scatter plot, (c) residuals by season, (d) SIRC compartmental dynamics during 2013–14 (highest latent drift season). Both models over-predict in high-drift years and under-predict in low-drift ones, consistent with the SIRC waning-rate parameterisation being too sensitive to extreme drift values.

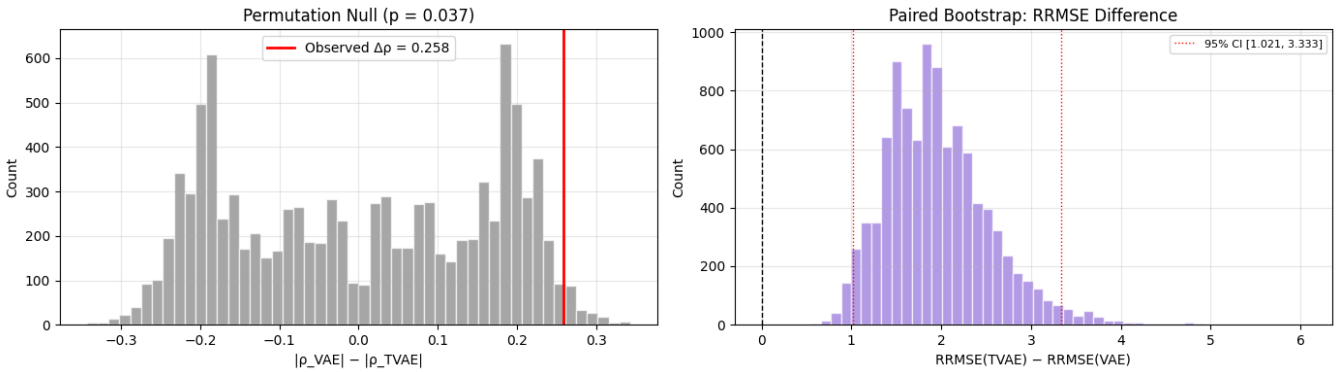

Figure 26: Left: permutation null distribution for $|\rho_{\text{VAE}}| - |\rho_{\text{TVAE}}|$; the observed difference of 0.258 falls at $p = 0.037$, indicating the TVAE achieves a significantly higher absolute Hamming correlation. Right: paired bootstrap distribution for RRMSE(TVAE)−RRMSE(VAE); the 95% CI $[1.021, 3.333]$ excludes zero, confirming the VAE is significantly *better* downstream.

D.4 SIRC parameter sensitivity

Figure 27 shows how predicted hospitalisation rates change as each of the three main calibrated parameters (IHR, S_0, $\delta_{\max}$) is varied over a grid while the others are held fixed. The output is most sensitive to IHR and S_0, which effectively scale the overall level; $\delta_{\max}$, conversely, controls the range of drift-driven variation but has secondary impact.

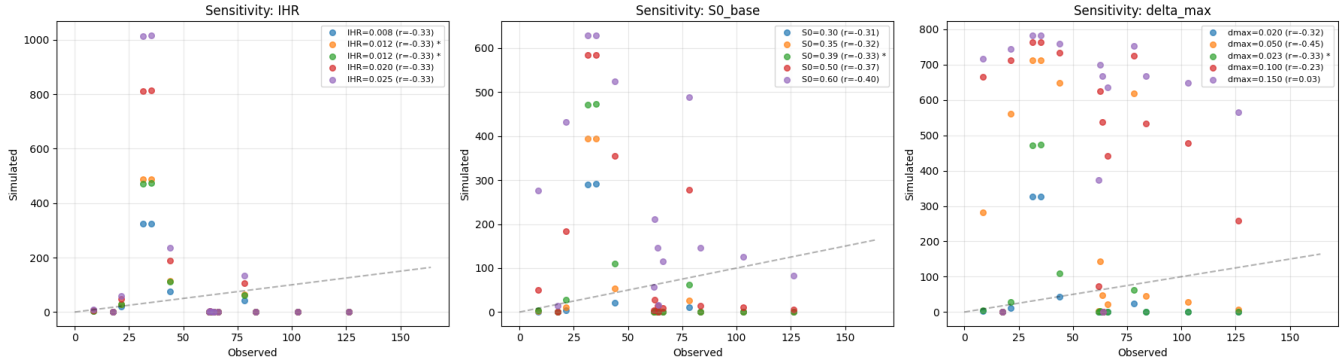

Figure 27: SIRC one-at-a-time sensitivity: each panel varies one calibrated parameter (IHR left, S_0 centre, $\delta_{\max}$ right) while the others are fixed at calibrated values. Hospitalisation output is most sensitive to IHR and S_0; $\delta_{\max}$ modulates the spread of season-to-season variation.

E Reproducibility Notes

E.1 Environment

I developed and executed the project on Google Colab Pro+ with an NVIDIA A100 GPU (40 GB). Key library versions used during the primary runs are listed below:

Package	Version
Python	3.12
PyTorch	2.5 (CUDA 12)
JAX	0.4.x
Diffrax	0.5.x
HuggingFace `datasets`	3.x
`umap-learn`	0.5.x
Optuna	4.x
MAFFT	7.x (system install)

I provide a `requirements.txt` in the repository root. Exact floating-point reproducibility is not guaranteed across different GPU hardware due to non-deterministic CUDA kernels in some PyTorch operations, even with fixed seeds.

E.2 Data access

All sequence and surveillance data are hosted on HuggingFace: `sidms/AML`. Relevant datasets:

- `H1N1/`, `H3N2/` — random-split global sequences (train/test Parquet)
- `H1N1_temporal/`, `H3N2_temporal/` — temporal-split sequences with season labels
- `H1N1_drift/`, `H3N2_drift/` — USA-filtered sequences for drift analysis
- `weights/` — saved model weights (`.pt` files)
- `results/` — SIRC simulation outputs and LOO-CV CSVs

I load weights automatically in each notebook via `hf_hub_download`. MAFFT must be installed separately (`apt-get install mafft` on Linux/Colab).

E.3 Notebook execution order

To reproduce all results from scratch, I recommend executing the notebooks in this order:

1. `seq_data.ipynb` — data download, alignment, upload to HuggingFace
2. `hyperparameter_VAE.ipynb` — Optuna search for Conv1D VAE
3. `hyperparameter_TVAE.ipynb` — Optuna search for TVAE
4. `vae_training_notebook.ipynb` — final VAE training with best config
5. `transformer_vae_training_notebook.ipynb` — final TVAE training
6. `cdc_surveillance_pipeline.ipynb` — CDC data ingestion and season summaries
7. `sirs_sirc_model.ipynb` — drift extraction, SIRC calibration, evaluation

Steps 2–5 are independent after step 1 and can be run in parallel. Step 7 depends on steps 4–5 (weights) and step 6 (surveillance data). Intermediate artefacts (weights, drift CSVs, season summaries) are uploaded to HuggingFace at the end of each notebook, so individual notebooks can be re-run in isolation once upstream artefacts exist.

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